Georgios-Chrysovalantis Chatzivantsidis

Physics Graduate | Data Science: Python, Statistics, SQL | Scientific Computing and Research

Thessaloniki, Greece | +30 694 783 0756 | hatzivantsidis@gmail.com | linkedin.com/in/georgios-chatzivantsidis-81a053110

PROFESSIONAL SUMMARY

Physics graduate (BSc, Aristotle University of Thessaloniki) moving into data science, with a quantitative and computational background. Wrote FORTRAN code for numerical simulation of asteroid orbits and co-authored a peer-reviewed *Astronomy & Astrophysics* paper based on large-scale numerical experiments on a supercomputer, plus a conference paper on stable chaos in asteroid motion. Taught university tutorials in Statistics, Probability and Theoretical Mechanics, including Statistics for an MSc in Computational Physics. Holds the Workearly Data Science Professional certification (Python, pandas, NumPy, Matplotlib, SQL, data modeling, big data analytics, AI-assisted data analysis), earned by passing exam projects and coding assignments, and completed a data analysis project on football data. Built a semantic search and RAG tool over 3,000 arXiv abstracts, evaluated against a BM25 baseline. Has built more than 10 apps with AI, plus several websites. Explains technical material clearly, from years of teaching students and adult learners.

TECHNICAL SKILLS

- **Programming:** Python (pandas, NumPy, Matplotlib); SQL; FORTRAN and Pascal (scientific computing); HTML, CSS, JavaScript; VS Code, Git
- **Statistics and mathematics:** statistics, probability, theoretical mechanics, nonlinear dynamics and chaos
- **Scientific computing:** numerical simulation and orbit integration; large-scale numerical experiments on a supercomputer
- **Data analysis:** data processing, data cleaning, data quality checks, ETL, data modeling, data warehouses, big data analytics, AI-assisted data analysis
- **AI tools:** ChatGPT, Claude, Gemini, DeepSeek and Qwen (regular use); embeddings, semantic search, RAG (Claude API)
- **Visualisation:** Matplotlib, Power BI (DAX, Power Query), Tableau, Excel; dashboards and reports
- **Technical writing:** TeX for scientific publishing; peer-reviewed publications
- **Office software:** Word, Excel, Access, PowerPoint (Cambridge IT Skills, Proficiency level)
- **Visual design:** Adobe Suite; book layout, cover and poster design

RESEARCH EXPERIENCE

Aristotle University of Thessaloniki, Laboratory of Astronomy - Thessaloniki, Greece

Researcher, Computational Celestial Mechanics

- Wrote FORTRAN code for numerical computation and simulation of asteroid orbits
- Co-authored a study on stable chaos in asteroid motion, presented at the international conference Asteroids, Comets, Meteors (ACM 2002, Berlin)
- Contributed to a study of whether terrestrial planets can survive in the habitable zones of five extrasolar systems; the work ran extensive numerical integrations on the Max Planck Institute for Gravitational Physics supercomputer and was published in *Astronomy & Astrophysics* (2004)

PUBLICATIONS

- Asghari, N., Broeg, C., Carone, L., ... Hadjivantsides, G., et al. (2004). Stability of terrestrial planets in the habitable zone of Gl 777 A, HD 72659, Gl 614, 47 UMa and HD 4208. *Astronomy & Astrophysics*, 426, 353-365.
- Varvoglis, H., Tsiganis, K., & Hadjivantsides, G. (2002). Stable chaos and local integrals of motion. *Proceedings of Asteroids, Comets, Meteors (ACM 2002)*, Berlin, ESA SP-500, 355-357.

TECHNICAL TRAINING AND CERTIFICATIONS

- **Data Science Professional** - Workearly (Credly certification, Feb 2026); Python for data analysis, SQL and databases, Power BI, Tableau, ETL, data modeling; passed exam projects and coding assignments
- **Data & Programming Summer School** - Workearly (Credly badge, Intermediate level, Feb 2026); Python, data analytics, AI, Excel and Power BI; passed exam projects
- **Python in Excel** - Workearly / Level Up, EU Digital Europe Programme (8 hours, Feb 2026)
- **Dedicated Web Application Developer (HTML, CSS, JavaScript)** - GlobalCert, ESYD-accredited (Oct 2023, valid to 2033)
- **Introduction to Programming with Python** - Aristotle University of Thessaloniki, Lifelong Learning (60 hours, 2 ECTS, Nov 2017); assessed through quizzes and coding mini-projects
- **International Diploma in IT Skills Proficiency** (Word, Excel, Access, PowerPoint, Outlook) - Cambridge International Examinations / Vellum (2014)
- **Chaos and Stability in Planetary Systems** - 3rd Helmholtz Summer School, Helmholtz Institute for Supercomputational Physics, Potsdam (Sep 2003; 80 hours of lectures, 50 hours of computing practicals; presented results)

PROJECTS

Semantic Paper Search - Personal Project (2026) | github.com/georgehadji/semantic-paper-search

- Built, with AI coding assistance, semantic search and cited question-answering (RAG) over 3,000 arXiv astro-ph.EP abstracts (Python, sentence-transformers embeddings, NumPy, Claude API, Streamlit, pytest)
- Compared it with a BM25 keyword baseline on 20 queries (176 LLM-judged query-paper pairs): precision@5 0.43 vs 0.40 overall, 0.22 vs 0.12 on plain-language queries

Apps and Websites - Personal Projects

- Built more than 10 apps with AI
- Built several websites

Football Data Analysis - Workearly Data & Programming Summer School (2026)

- Final data analysis project on football data, including xG and other performance indicators
- Cleaned the data and ran data quality checks
- Processed and visualised the data with Python, Excel and Power BI, and built interactive dashboards and reports
- Presented the findings

WORK EXPERIENCE

Kyriakidis Publications - Thessaloniki, Greece

Graphic Designer

November 2017 - present

- Lay out books for a publishing house
- Design book covers and posters
- Work in Adobe Suite
- Manage own workload and meet publishing deadlines

SAEK Epanomis (Public Vocational Training School) - Epanomi, Greece

Adult Trainer (hourly contract)

March 2024 - June 2024

- Taught "Special Topics in Physics" (theory and lab) to adult vocational trainees, 36 teaching hours

Directorate of Secondary Education of East Attica - Avlona, Greece

Substitute Physics Teacher, 2nd Gymnasium & Lyceum Classes of Avlona

October 2008 - June 2012

- Taught Physics for four school years at a school operating inside the Avlona Special Juvenile Detention Centre (30 months and 24 days of recognised service)

Thessaloniki Planetarium (Tsampouras D. & Co OE) - Thessaloniki, Greece

Tour Guide, Mobile Planetarium

March 2008 - October 2008 (full-time); January 2003 - December 2003 (part-time)

- Led planetarium tours for primary and secondary school groups

Aristotle University of Thessaloniki, Department of Physics - Thessaloniki, Greece

Teaching Assistant ("Heraclitus" Research Scholarship)

February 2004 - June 2005

- Taught "Statistics" tutorials in the MSc programme in Computational Physics (Feb-Jun 2005)
- Taught "Statistics & Probability" (Feb-Jun 2004) and "Theoretical Mechanics I" (Sep 2004-Jan 2005) tutorials for undergraduates

EDUCATION

BSc in Physics - Aristotle University of Thessaloniki

1994 - 2000 | Grade 6.7/10 ("Very Good") | Physics studies began at the University of Crete (1992-1994)

Vocational Diploma in Print and Electronic Media Graphic Design - IEK AKMI, Thessaloniki

Training completed July 2019 | 1,173 hours

ADDITIONAL TRAINING

- Extrasolar Planetary Systems - Wilhelm and Else Heraeus Summer School, Bad Honnef, Germany (2005)
- Complexity and Chaotic Dynamics of Nonlinear Systems - Greek summer schools, Chalkida (2003) and Patras (2001)
- Particle Physics Training Programme for Teachers - CERN, Geneva (2010)
- Training of Adult Educators - University of Macedonia (160 hours, 2023-2024)

LANGUAGES AND MEMBERSHIPS

- **Greek:** native | **English:** proficient - Cambridge Certificate of Proficiency in English (CPE, 2001)
- Association of Greek Physicists; Hellenic Astronomical Society (Hel.A.S.)